



5 A continuous random variable X has probability density function f given by

$$f(x) = \begin{cases} \frac{1}{16}\sqrt{x} & 0 \leq x < 4, \\ \frac{1}{k\sqrt{x}} & 4 \leq x \leq 9, \\ 0 & \text{otherwise,} \end{cases}$$

where k is a constant.

(a) Show that $k = 3$.

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(b) Find the median value of X .

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The random variable Y is defined by $Y = \sqrt{X}$.

- (c) Find the probability density function of Y .

[5]

[illegible]